

Measuring Face Tell: An End-to-End Facial Expression Pipeline, and Two Problems Found in Its Data

A technical report on what replicated, what failed, and what was contaminated

Jawad Hassan

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github.com/jawadhasuna/Face-Tell · facetell.vercel.app

ABSTRACT

We document a complete facial expression recognition (FER) pipeline built on a single consumer GPU and deployed to the browser, and we report every measurement taken along the way, including the ones that contradicted expectation. Two representations are compared on identical data: MediaPipe's 52 facial-muscle activations with a gradient-boosted classifier, and raw pixels with a fine-tuned EfficientNet-B0. Pixels win by 24.6 points of macro recall, with the gain concentrated almost entirely in the three classes the muscle representation fails at (fear +44, sad +42, disgust +41).

The more consequential results are negative. Evaluating across datasets rather than within one, roughly half of measured performance disappears. Investigating that gap surfaced two data problems in widely-downloaded dataset mirrors: 5,130 byte-identical images shared between an AffectNet mirror and a RAF-DB mirror, which silently inflates any cross-corpus evaluation performed on the pair; and a semantic disagreement about the label *happy* severe enough that a model trained on one scored 0.1% on the other's happy faces. Merging the datasets raised that figure to 94% with no architectural change. We further report a hyperparameter sweep, an INT8 quantisation that cost 10.5 accuracy points rather than the customary one, and a LoRA fine-tune that renders muscle measurements as English sentences.

KEYWORDS — facial expression recognition, cross-corpus generalisation, dataset contamination, transfer learning, blendshapes, LoRA, on-device inference.

1 Scope of the claims

This report is largely a replication. Stating that first makes the remainder easier to read. Transfer learning outperforming hand-crafted features, cross-corpus degradation in FER, and augmentation disproportionately helping rare classes are all established results; they are reproduced here with measurements because reproducing a known result on one's own data is how one learns whether the pipeline is correct.

Two findings are, to our knowledge, not previously documented: the dataset contamination and the quantified *happy* semantic mismatch, both in §5. Three results are negative — a standard technique that did not work here. Each section is tagged accordingly.

2 Selecting data by measurement

REPLICATION The blendshape representation depends on MediaPipe locating a face; where it cannot, the image contributes nothing. Detection rate was therefore measured before any dataset was downloaded in full.

TABLE 1. MediaPipe face detection rate, 120 images per condition.

Dataset	Resolution	DetectedOutcome
FER-2013	48 × 48	—could not load (script-based repository)
RAF-DB	100 × 100	85.8%usable
RAF-DB (upscaled)	200 × 200	85.0%no improvement
AffectNet-HQ	1547 × 1547	97.5%selected

The third row is the informative one: upscaling RAF-DB by 2× changed detection by −0.8%. Interpolation adds no information, so a small original remains a small original. This eliminated FER-2013 — the dataset most FER tutorials adopt — on evidence rather than preference. A follow-up measurement found detection and latency both flat from 256 px to 1547 px (97%, 10–14 ms), because MediaPipe resizes internally; inference was never the bottleneck.

NEGATIVE **Transfer strategy.** Streaming the datasets to avoid disk usage ran at 3.4 images/s, and a 500-image shuffle buffer holding decoded 1547 px frames grew to 11 GB of resident memory. Downloading first and processing locally ran at 85 images/s — a 25× difference. Disk was the inexpensive resource.

3 Baseline: 52 activations in place of pixels

REPLICATION MediaPipe FaceLandmarker emits 52 blendshapes — ARKit-style muscle activations (mouthSmileLeft, browDownRight, jawOpen) in [0,1] — at 26 FPS on CPU. A 640×480 frame comprises 921,600 values; the blendshape vector comprises 52, with subject identity already factored out upstream. Four classifiers were fitted per dataset; gradient boosting won both times.

TABLE 2. Blendshape baselines, held-out stratified split. Accuracy (%), with macro recall of the winning model.

Dataset	n	Logistic	Forest	Boosting	MLP	Macro
RAF-DB	18,329	63.1	71.4	73.8	71.5	55.6
AffectNet	27,699	51.6	55.1	56.3	53.7	53.6

RAF-DB records 73.8% accuracy against 55.6% macro recall — an 18-point divergence. Fifty-seven percent of its images are *happy* or *neutral*, so a model may appear competent while remaining useless on the remainder. AffectNet is near-balanced (1.9:1 against RAF-DB's 17:1) and its two figures nearly agree. Accuracy was misleading on one dataset and honest on the other; macro recall was honest on both, and is used throughout this report.

3.1 Partitioning dominates model choice

On 2,100 webcam frames of a single subject, a random split yielded 100.0% accuracy and a temporal split 91.8%. Consecutive video frames are near-duplicates, and a random split distributes them across both partitions. The 8.2-point difference is an artefact of partitioning alone.

4 Transfer learning: where pixels win

REPLICATION EfficientNet-B0 pretrained on ImageNet, its 1000-class head replaced by a 7-class head. Training proceeds in two stages: two epochs with the backbone frozen, so that the randomly initialised head does not propagate large gradients into pretrained features, followed by all 4.0M parameters unfrozen at a reduced learning rate under cosine decay. bfloat16 throughout; peak memory 2.26 GB of 8 GB. Validation accuracy held at 52% across the two frozen epochs and rose 23 points in the single epoch following unfreezing.

TABLE 3. Per-class recall (%) on RAF-DB. Both models trained and evaluated on the identical split.

Class	Blendshapes	CNN	Δ	Train n
neutral	94	100	+6	5,130
happy	85	93	+8	5,197
angry	68	76	+8	3,343
surprised	65	88	+23	1,466
sad	41	83	+42	2,123
disgusted	21	62	+41	765
fear	16	60	+44	305
macro recall	55.6	80.2	+24.6	

The distribution of the gain is the result, not its magnitude. Classes the blendshapes already handled improved by approximately 8 points; the three at which they failed improved by 41–44. Disgust and anger share lowered brows and a tightened mouth; fear and surprise share widened eyes and an opened jaw. The features distinguishing each pair — creasing at the nose bridge, the precise curl of the upper lip — do not survive compression into 52 scalars, but do survive in pixels. Note also that fear improved by 44 points from only 305 training examples, which indicates the limitation was representational rather than purely one of sample size.

5 Cross-dataset evaluation

REPLICATION **NEW** A held-out split establishes only that a model generalises to further photographs from the same collection. The stricter test is a different collection.

TABLE 4. Identical models evaluated on their own held-out split and on a dataset never seen in training.

Model	Own split	Other dataset	Retained
Blendshapes + boosting (RAF-DB)	73.8	33.4	45%
EfficientNet-B0 (AffectNet)	75.3	≈44	58%

The blendshape model did not degrade gracefully; it collapsed, predicting *neutral* for nearly all inputs (99.4% recall on neutral, single digits elsewhere). The mechanism is that MediaPipe reports weaker activations from RAF-DB's soft 100 px originals than from AffectNet's sharp ones, so every learned decision threshold falls in the wrong place. Blendshapes are person-independent; they are *not* capture-independent. This distinction does not appear to be widely stated and is material for any work treating blendshapes as a normalised representation.

FINDING 1 — 5,130 IMAGES ARE SHARED BETWEEN THE TWO DATASETS

Mean blendshape values for *neutral* agreed to three decimal places across both datasets, and both neutral classes contained exactly 5,130 rows. Hashing the extracted images confirmed 5,130 byte-identical files shared between `Pirol17/affectnethq` and `deanngkl/raf-db-7emotions` — the entire neutral class of one, duplicated in the other. No other class overlaps at all.

The consequence is direct: the AffectNet-trained model's apparent 96% cross-dataset transfer on *neutral* was recognition of images it had trained on. Excluding the contaminated class, honest cross-dataset macro recall is 42.8% rather than the 50.4% the naive evaluation reports. Any cross-corpus result computed on this pair of mirrors is inflated by an amount that depends on how heavily neutral is weighted.

FINDING 2 — THE TWO DATASETS DISAGREE ON THE MEANING OF *HAPPY*

Among the uncontaminated classes one result was anomalous: the AffectNet-trained CNN scored 0.1% on RAF-DB's happy faces — 5 correct of 5,197 — while scoring 60–68% on angry, fear and sad. A uniform domain shift does not destroy exactly one class.

Two hypotheses were tested and rejected. Labels are not transposed: both folders visibly contain smiling faces. Resolution is not the cause: degrading AffectNet images to RAF-DB's effective 100 px left happy recall at 95%.

The cause is semantic. AffectNet's *happy* is predominantly a mild, closed-mouth adult smile; RAF-DB's is a wide, toothy laugh, frequently a child's. Measured on the blendshapes, `mouthStretch` is 2.8× higher in RAF-DB's happy class. To a model trained on gentle smiles, bared teeth with a stretched mouth read as fear, disgust or anger; its confidences on those misclassifications were 0.34–0.76, indicating uncertainty rather than confident error.

5.1 The remedy is data, not architecture

Merging both datasets with hash-deduplication (40,665 images retained, 5,363 duplicates discarded) and retraining under identical hyperparameters:

TABLE 5. Merged model, per-class recall (%) on the held-out test split, decomposed by source dataset.

Class	AffectNet-origin	RAF-DB-origin
happy	93	94
angry	65	85
sad	66	89
surprised	66	89
fear	69	56
disgusted	65	61
accuracy / macro	76.0 / 73.5	87.2 / 79.0

Happy rose from 0.1% to 94%. The architecture, hyperparameters and epoch count were unchanged; the sole intervention was exposing the model to both definitions of the word. The merged model additionally exceeds the AffectNet specialist on AffectNet's own test split (76.0 against 75.3) while conceding three points on RAF-DB.

A further data-quality observation: 54 images within AffectNet carry two distinct labels — the same photograph filed as both *angry* and *disgusted*, or *disgusted* and *fear*. A dataset that disagrees with itself is a useful reminder of why published AffectNet results plateau near 65%, and why human annotators agree only 60–65% of the time.

6 Hyperparameter sweep

REPLICATION One factor varied at a time against a fixed reference configuration (EfficientNet-B0, 12 epochs, batch 48, lr 3e-4, medium augmentation, class-weighted loss). Twelve runs, all logged.

TABLE 6. Sweep results on RAF-DB, ranked by macro recall. The final column is the train–validation accuracy gap at the last epoch.

Configuration	Test acc.	Macro	Gap
aug = strong	88.8	81.8	3.0
lr = 1e-3	88.5	80.7	6.4
aug = medium (reference)	88.4	80.2	7.2
aug = light	87.2	79.2	8.7
lr = 1e-4	84.8	79.0	4.5
strong + 1e-3, 30 epochs	90.6	82.7	6.9

Learning rate behaves as a threshold, not a dial. 3e-4 and 1e-3 differ by three images of 2,745. At this sample size the 95% confidence interval is ± 1.2 points, so the difference is not measurable. Reducing to 1e-4 costs 102 images, which is outside the interval — but its validation curve was still ascending at the final epoch, so 1e-4 is *undertrained at 12 epochs* rather than inferior. Reading only the final figure would have produced the wrong conclusion.

Strong augmentation is justified by the rare classes. The accuracy difference between medium and strong (0.4 points, 11 images) lies within noise. What lies outside it is that strong wins on four of seven classes and on all three difficult ones — disgust 62→66, fear 60→64, sad 83→85. A single fluctuation moves one figure; this moved a pattern. The train–validation gap simultaneously fell from 8.7 to 3.0.

Combining both findings produced the best single-dataset result reported here, 90.6% accuracy and 82.7% macro recall, converging at epoch 25. Published RAF-DB results are approximately 88–90%, so this is at the level of the literature and not beyond it.

Two runs of an identical configuration produced identical figures to four decimal places, which eliminates run-to-run variance as a confound when comparing configurations. It does not eliminate test-set sampling error, which the confidence interval above accounts for.

7 Deployment

NEGATIVE The model was exported to ONNX, agreeing with PyTorch to 1×10^{-6} , and quantised to INT8 using per-channel symmetric post-training quantisation calibrated on 200 real images. The conventional expectation is approximately 4× compression for about one point of accuracy.

TABLE 7. ONNX variants evaluated on the full 2,745-image test set, CPU.

Variant	Size	Accuracy	ms / image
float32	16.1 MB	90.6	5.4
int8 (PTQ, per-channel)	4.9 MB	77.8	6.1

INT8 cost 10.5 accuracy points and delivered no speedup. EfficientNet quantises poorly: depthwise-separable convolutions exhibit widely varying per-channel dynamic ranges, and squeeze-and-excitation blocks with smooth activations are sensitive to rounding. A residual architecture would likely have quantised cleanly. float32 was deployed. The general point is that the customary figure is architecture-dependent and worth measuring rather than citing.

7.1 Crop geometry

The live pipeline crops a face from the camera frame using MediaPipe landmarks before classification. An initial margin of 0.25, chosen by intuition, visibly degraded accuracy. Measurement across 199 training images established that the training frame is 0.97× the landmark bounding box: the 478 landmarks already extend to the hairline, so the box *is* the head. The correct margin was 0.00 and the intuition had been inverted.

7.2 Model selection is a function of deployment conditions

The merged model generalises better by every offline measure. Evaluated side by side on an actual webcam, the RAF-DB model performed noticeably better, and the reason is legible: a 640×480 frame in dim lighting, cropped to roughly 200 px, resembles RAF-DB's soft upscaled originals considerably more than AffectNet's sharp 1547 px photography. The merged model expended half its capacity on a domain the application never encounters. On higher-quality capture the ranking would likely reverse.

8 A natural-language describer

REPLICATION A classifier emits a single word. To enable the application to explain itself, Llama-3.2-3B-Instruct was LoRA fine-tuned to render the same muscle measurements as a sentence. No image-to-sentence dataset for facial expressions exists, so one was generated from measurements already obtained; every clause traces to a numeric value. Intensity bands (*faintly*, *clearly*, *strongly*, *very strongly*) were set from the observed value distribution across both datasets: 69% of readings fall below 0.05, p90 = 0.32, p95 = 0.53, p99 = 0.84.

Two curation passes, both prompted by inspecting generated output:

- Gaze direction indicates where attention is directed, not what is felt. `eyeLook*` signals were demoted so that they surface only when no other signal is active.
- 6,620 rows (14.4%) were discarded where the measurement flatly contradicted the dataset label — for instance `mouthSmile 0.95` on an image labelled *fear*. One of the two is wrong, and training on the pair teaches contradiction.

3,990 class-balanced examples, 28 minutes on an RTX 3050, peak 3.2 GB. 24.3M of 3.24B parameters trained (0.75%). Loss fell 3.25 → 0.29 and converged by step 100, indicating that roughly 1,600 examples would have sufficed for what is essentially format learning.

input	mouthSmile 0.88, eyeSquint 0.61, cheekSquint 0.34
before	"Based on the facial muscle activations, this person is making a slight, subtle smile ... the expression appears to be a gentle, slightly annoyed"
after	"The mouth corners pulled up very strongly, the eyes narrowed strongly."

LISTING 1. Held-out probe before and after fine-tuning. The base model characterised an activation of 0.88 as "slight"; on another probe it described 0.33 as a "high activation" and 0.41 as "lower".

THE MODEL DESCRIBES WELL AND INFERS POORLY

Given browDown 0.74, mouthPress 0.48, noseSneer 0.21 — an anger signature — the model produced an accurate muscle description and then concluded "the overall read is sad." The CNN performs that classification at 90.6%, so the deployed system splits the generated text on its final sentence, displays the description, and takes the label from the CNN. The language model's own verdict is returned in the response and never displayed. Placing the classifier's answer in the prompt would be the better design and remains future work.

9 System, limitations, reproducibility

TABLE 8. Deployed pipeline. Only the describer leaves the device, and it receives 52 numeric values, never an image.

Stage	Location	Model	Latency
Face tracking	Browser (WASM)	MediaPipe FaceLandmarker	≈15 ms
Expression	Browser (WASM, 4 threads)	EfficientNet-B0 / ONNX fp32	29 ms
Nervousness	Browser	none — temporal arithmetic	<1 ms
Description	Cloud T4, scale-to-zero	Llama-3.2-3B + LoRA r=16	3.1 s warm

Multi-threaded WASM requires `SharedArrayBuffer`, which requires COOP and COEP response headers. Without them `onnxruntime-web` silently falls back to a single thread, a roughly 3× latency penalty that produces no error.

9.1 Nervousness estimation

Nervousness is not an expression; no static face denotes it. It is estimated as a weighted deviation from a per-subject 30-second baseline across blink rate (40%), gaze instability (25%), expression churn (20%) and sustained tension classes (15%). Per-subject calibration is not optional: MediaPipe's blink signal peaked at 0.706 on the test camera, so a textbook 0.9 threshold detects no blinks whatsoever. Two calibration runs on the same subject measured resting rates of 9 and 22 blinks/minute, so the score is comparable within a session but not across sessions. Measured values: 15–20 at rest, 70 while performing anxiety.

9.2 Limitations

- Cross-dataset macro recall remains approximately 44%. Two datasets are insufficient, and no claim of general-purpose accuracy is made.
- `Piro17/affectnethq` is a curated 27,823-image subset, not the ≈440k AffectNet benchmark; its figures are not comparable to published AffectNet results. RAF-DB figures are approximately comparable.
- No identity labels are available, so a subject may appear in both the training and test partitions of a single dataset. Within-dataset figures may be slightly optimistic.
- Both datasets are third-party mirrors of collections normally distributed under signed academic licence. They comprise photographs of real people who did not consent to inclusion in a training set. No face data is redistributed with this work.

- Finding 1 was established by hashing images after a deterministic resize. Hashing the source archives directly, and confirming the overlap persists in current revisions of both repositories, would strengthen it.

9.3 Reproducibility

All twelve training runs are logged as JSON containing the complete configuration, per-epoch history and confusion matrices. The pipeline is deterministic under a fixed seed: two runs of one configuration agreed to four decimal places. Code, logs and the deployed application are available at the addresses given beneath the title.

Hardware and software. NVIDIA RTX 3050 (8 GB, 9.6 TFLOPS fp16), Windows 11, Python 3.12, PyTorch 2.11+cu128, MediaPipe 1.0.1, scikit-learn 1.9, onnxruntime-web 1.27, Unsloth 2026.8.19. Total compute for every result reported here is under six GPU-hours.

Datasets. RAF-DB (Li & Deng, Beijing University of Posts and Telecommunications) and AffectNet (Mollahosseini, Hasani & Mahoor, University of Denver), accessed as third-party HuggingFace mirrors `deanngkl/raf-db-7emotions` and `Pirol7/affectnethq`.